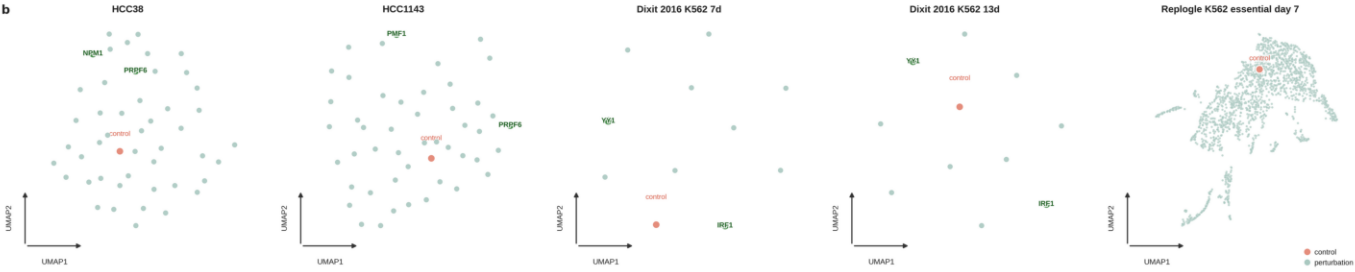


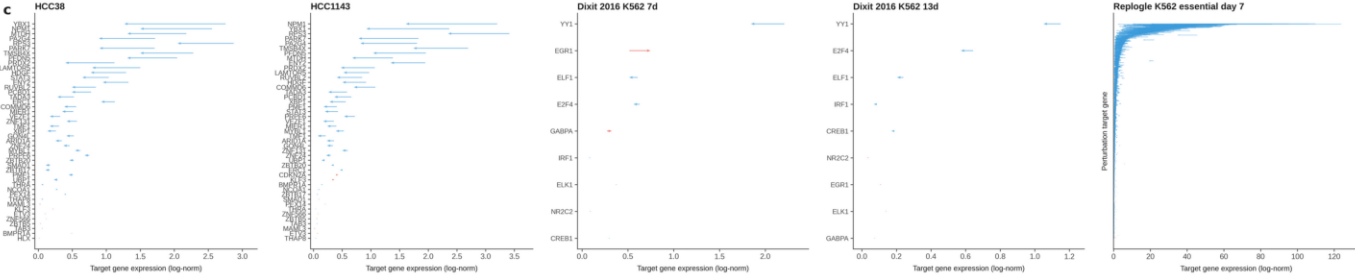
a

Dataset	Size	Benchmark use
Perturbation datasets		
HCC38	36,601 genes x 14,175 cells	47 targets 1,666 controls
HCC1143	36,601 genes x 11,405 cells	47 targets 1,325 controls
Dixit 2016 K562 7d	20,824 genes x 28,034 cells	10 targets 5,381 controls
Dixit 2016 K562 13d	20,824 genes x 15,849 cells	10 targets 3,491 controls
Replogle K562 essential day 7	8,563 genes x 310,385 cells	2057 targets 10,691 controls
Endpoint datasets		
DepMap CRISPR dependency (primary endpoint)	1,186 cell lines x 18,435 genes	HCC38 47 / HCC1143 48 / K562 7d 10 / K562 13d 10 / Replogle K562 1,882
DEMETER2 RNAi (sensitivity endpoint)	701 mapped cell lines x 17,309 genes	HCC38 48 / HCC1143 48 / K562 7d 9 / K562 13d 9 / Replogle K562 1,882

b

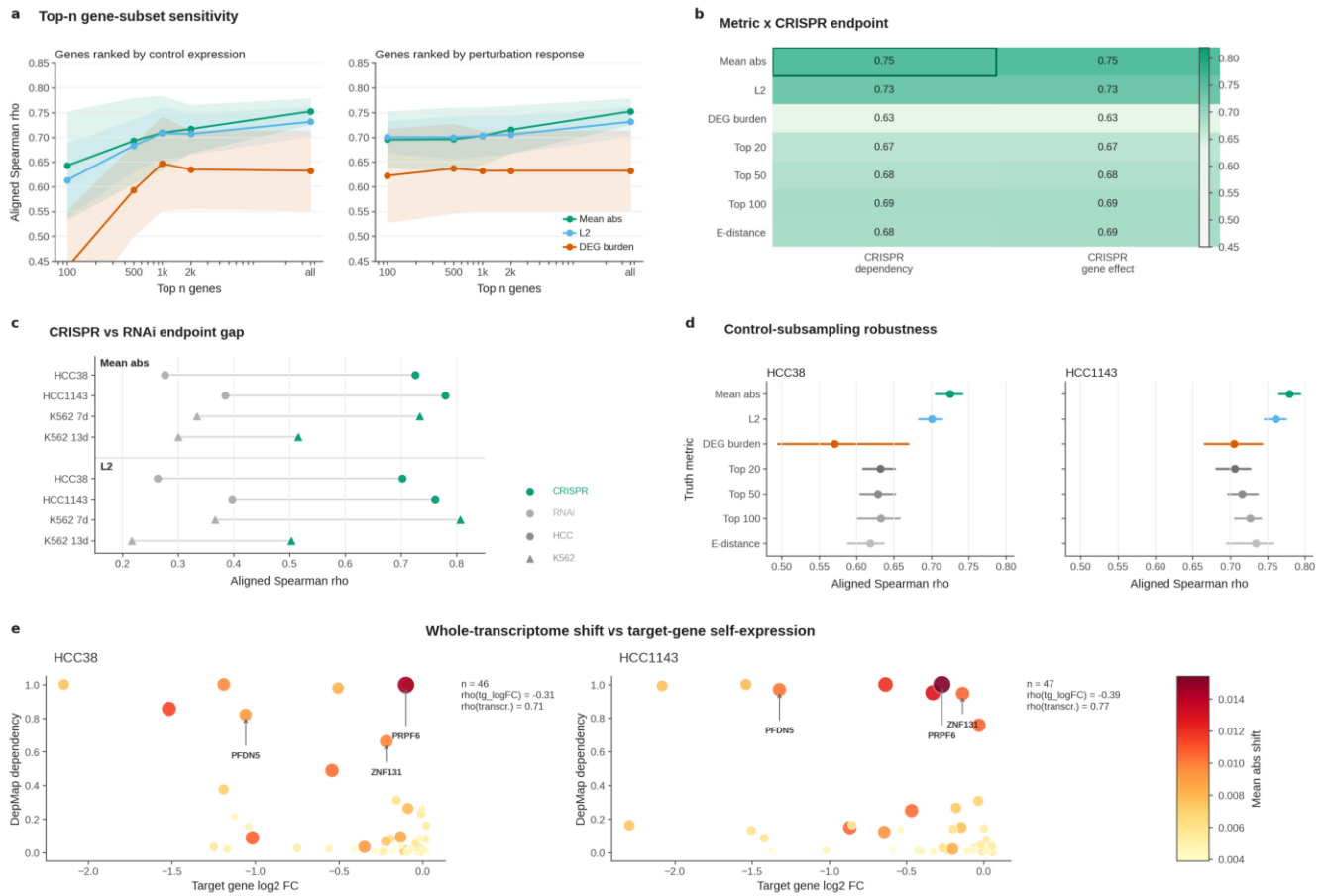


c



Extended Data Fig. 1 | Dataset familiarization.

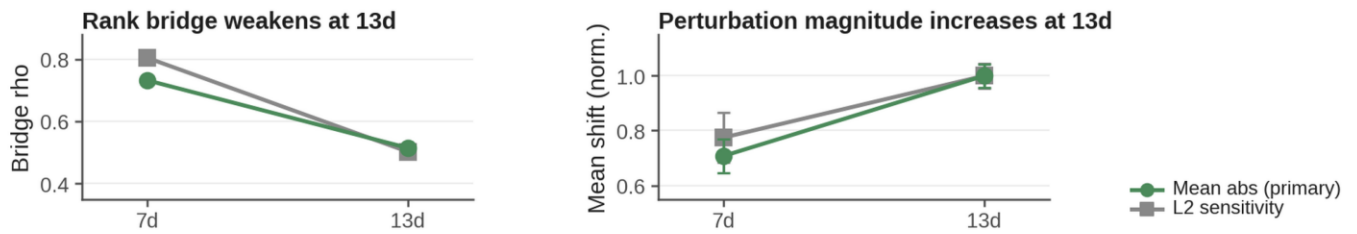
a, Dataset overview for five perturbation-expression contexts and two endpoint resources, summarizing cell-line identity, size and benchmark use. b, UMAPs of perturbation-level mean profiles for HCC38, HCC1143, Dixit K562 7d, Dixit K562 13d and Replogle K562 essential day 7; the matched control aggregate is marked. c, Target-gene expression changes for the same five contexts, with arrows from control expression to post-perturbation expression.



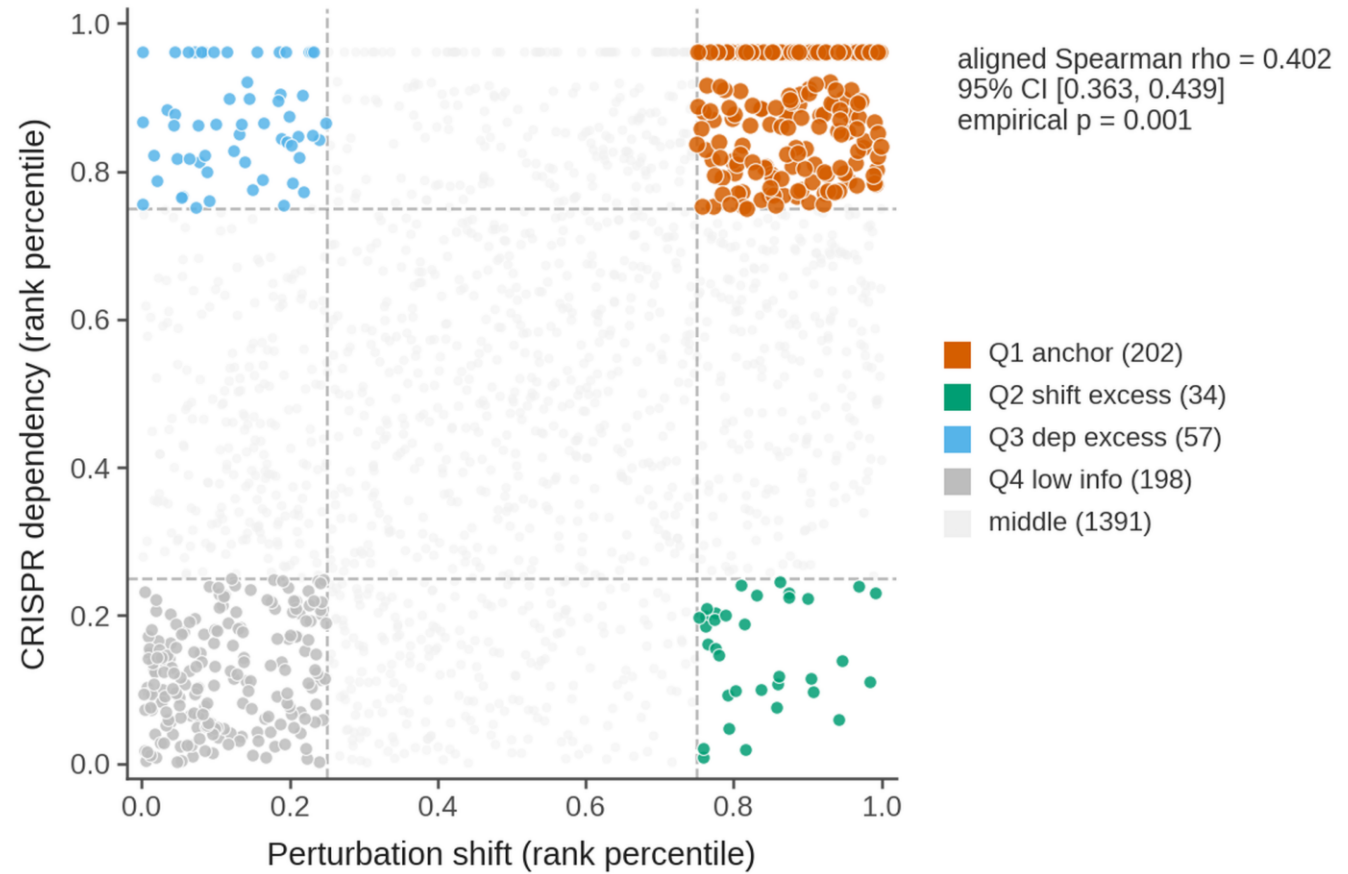
Extended Data Fig. 2 | Metric robustness.

a, Top-n gene-subset sensitivity recomputes aligned Spearman correlation after ranking genes by control expression or perturbation-response magnitude. b, Metric and CRISPR endpoint heat map marks the retained primary bridge metric and endpoint. c, Endpoint sensitivity compares CRISPR dependency with RNAi DEMETER2 across primary HCC and supplementary K562 contexts. d, Control-subsampling robustness shows mean and 2.5th-97.5th percentile ranges across repeated control-cell subsamples. e, Whole-transcriptome shift, rather than target-gene self-expression, carries the fitness bridge.

a K562 temporal stratification (7d vs 13d)

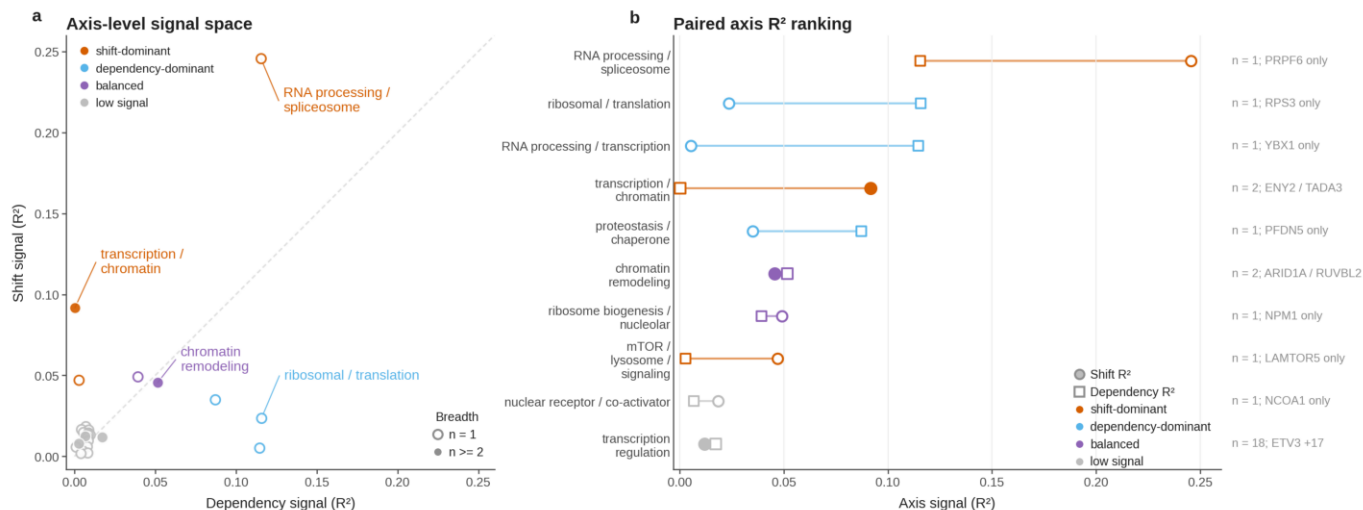


b Large-scale perturbation-fitness bridge confirmation



Extended Data Fig. 3 | K562 and Replogle bridge support.

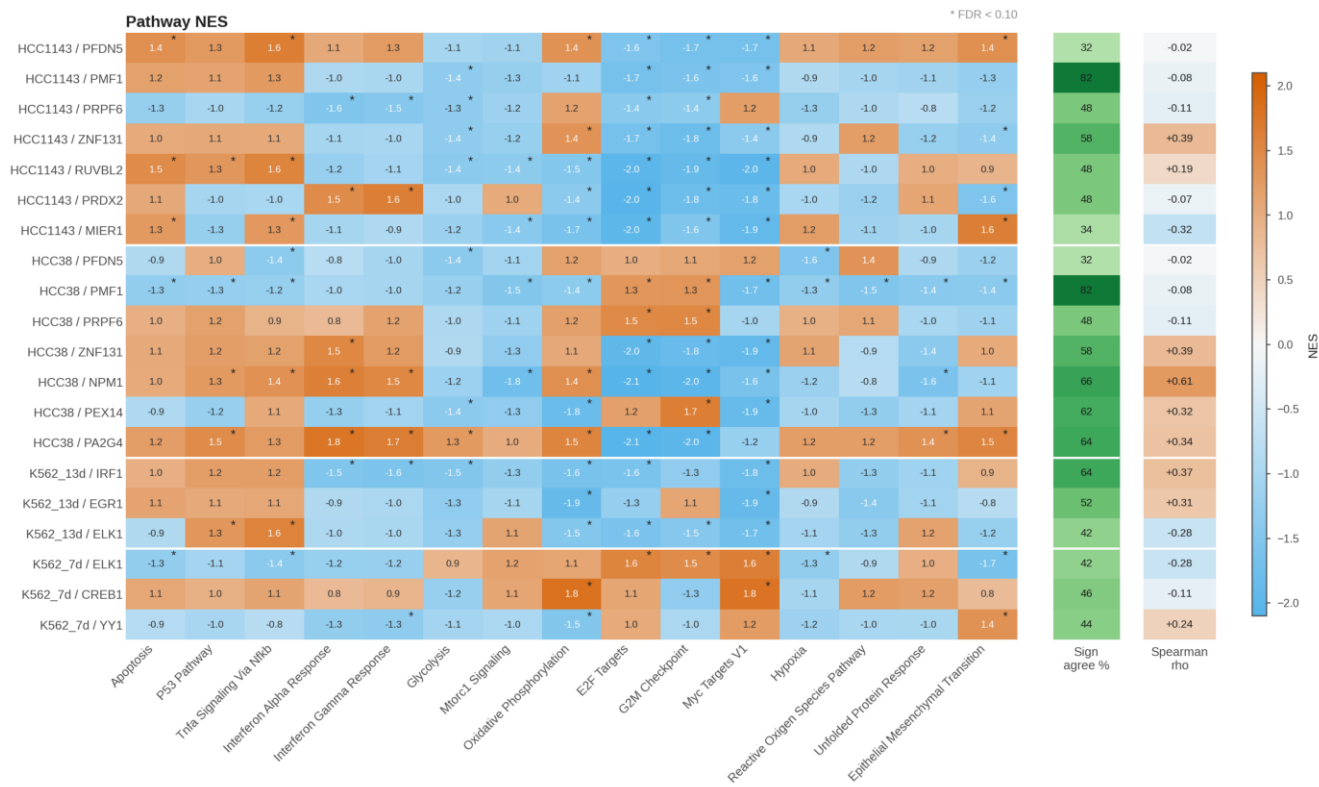
a, K562 temporal bridge-magnitude dissociation shows aligned Spearman correlation with CRISPR DepMap dependency and mean perturbation-shift magnitude at 7d and 13d; 13d has larger magnitude but weaker rank alignment. b, Replogle K562 essential day 7 bridge test plots 1,882 matched CRISPRi targets as rank percentiles for perturbation shift and dependency, with 25/75 percentile cutoffs, Q1-Q4 category counts and aligned Spearman rho = 0.402.



Extended Data Fig. 4 | Axis-level signal space.

a, Axis-level signal space compares dependency signal with transcriptomic shift signal across annotated axes; color denotes descriptive signal profile rather than claim tier. b, Paired axis R-squared ranking displays axes with the largest signal in either dimension on a common scale, supporting an audit-style view of partial axis-level structure.

a Exploratory pathway-level summaries of target-control perturbation responses



Extended Data Fig. 5 | Pathway-response polarity.

a, Pathway normalized enrichment score heat map across pre-specified display targets in HCC38, HCC1143, K562 7d and K562 13d. Rows show selected anchors and high-variance response examples; columns show the Hallmark response panel; asterisks mark FDR < 0.10. Right-hand summaries report same-target partner-context pathway Spearman correlation and sign agreement, treating pathway polarity as exploratory context rather than a benchmark-defining endpoint.